
Topic 10

Validity

Overview

- Basic knowledge
- Construct validity
- External validity
- Internal validity
- Statistical validity

Basic knowledge

Experiment

Instruction:

You will be presented with a list of colored words. It'll be your task to sequentially name the colors of the words. Please be sure to clock the time you need to name all colors in the list.

STROOP TASK (STROOP, 1935)

Name the *color* of the letters!

XXXXX

STROOP TASK

Do it again: Name the *color* of the
letters!

RED

Basic knowledge

green

red

blue

red

brown

blue

Basic knowledge

zyp

htklt

ygtde

gtopr

frangf

hyfop

Basic knowledge

blue

brown

red

bleu

red

green

Basic knowledge

Condition

Incongruent

Neutral

Congruent

blue

zyp

green

brown

htklt

red

red

ygtde

blue

blue

gtopr

red

red

frangf

brown

green

hyfop

blue

Time_i

Time_n

Time_c

Time_i > Time_n > Time_c

Basic knowledge

- ***Independent*** variables
 - Manipulated variables
 - *e.g., Condition - Congruence (incongruent; neutral; congruent)*
- ***Dependent*** variables
 - Measured variables
 - *e.g., Time or RT*

Basic knowledge



Confounding variables

- Variables that change simultaneously with the independent variables
- *e.g., time of testing, order*

• ***Control*** variables (constants)

- Variables that are being held constant on purpose
- *e.g., number of words, colors etc.*

• Random error

- *e.g., noise during testing*

Basic knowledge

- Anxiety as a variable of interest



Task: 'Name the ink colour as fast as possible'

- BUT anxiety is an abstract concept*

Basic knowledge

- ***Conceptual*** variable (construct)
 - a variable stated at an abstract (theoretical) level
 - *e.g., anxiety*
- ***Operational*** variable (operational definition)
 - The specific way in which a construct is manipulated or measured in a study
 - "How did the researcher manipulate or measure the construct?"
 - Assigning numbers is a good start
 - *e.g., test score -> control or anxiety group*

“People who sit within two tables of the bartender have three more alcoholic drinks, on average, than those who sit three tables away.”

What are the variables in this claim?



0

0

0

- a. sitting within two tables of the bartender versus sitting three tables away
- b. sitting within two tables of the bartender
- c. how far you sit from the bartender and how many drinks you have
- d. having three extra alcoholic drinks

Basic knowledge

- Goal of science
 - To find good justification for the belief that a specific claim or statement (hypothesis) is true or false

Goals of Psychology

1. To **describe** behavior and mental processes
2. To **predict** behavior and mental processes
3. To **explain** and **understand** behavior and mental processes
4. To **influence/control** behavior and mental processes

Basic knowledge

- Three claims (statements)
 - Frequency claims
 - focus on one variable
 - *descriptive* research

CLAIM TYPE	SAMPLE HEADLINES
Frequency claims	4 in 10 teens admit to texting while driving 42% of Europeans never exercise Middle school kids see 2-4 alcohol ads a day

Basic knowledge

- Three claims (statements)
 - Association claims
 - focus on the relationship between at least two variables
 - Association
 - Correlation
 - Covariation

CLAIM TYPE	SAMPLE HEADLINES
Association claims	Single people eat fewer vegetables Angry Twitter communities linked to heart deaths Girls more likely to be compulsive texters Suffering a concussion could triple the risk of suicide

Which of the following is an example of an association claim?

- a. 41% of people surveyed reported that they were having a good day.”
- b. “Guzzling a 20-ounce bottle of soda ramps up cellular aging.”
- c. “People who sit within two tables of the bartender have three more alcoholic drinks, on average, than those who sit three tables away.”
- d. “Viewing a recent conflict as it would look one year in the future led to increased feelings of forgiveness.”



they were

0

“Guzzling a 20-ounce bottle of soda ramps up cellular aging.”

0

“People who sit within two tables of the bartender have three more alcoholic drinks, on average, than those who sit three tables away.”

0

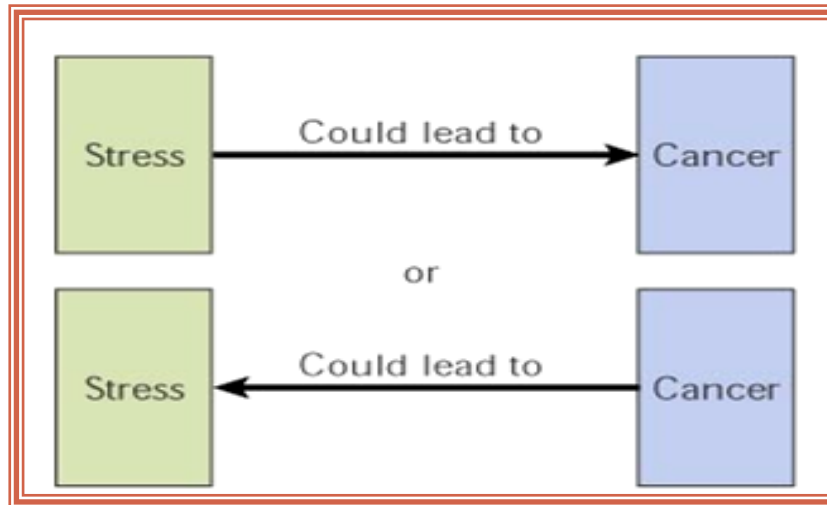
“Viewing a recent conflict as it would look one year in the future led to increased feelings of forgiveness.”



Goals of Psychology

To **predict** behaviors, thoughts, feelings, change and so on...

- Mainly by way of assessing the relationship between two or more variables



- **Correlations do not permit inferring causality**



Basic knowledge

- Three claims (statements)
 - Causal claims
 - focus on causation:
 - A change in one variable is responsible for changing the value of another variable
 - ***Experimental*** research

CLAIM TYPE	SAMPLE HEADLINES
Causal claims	Music lessons enhance IQ Babysitting may prime brain for parenting Family meals curb eating disorders Why sleep deprivation makes you crabby

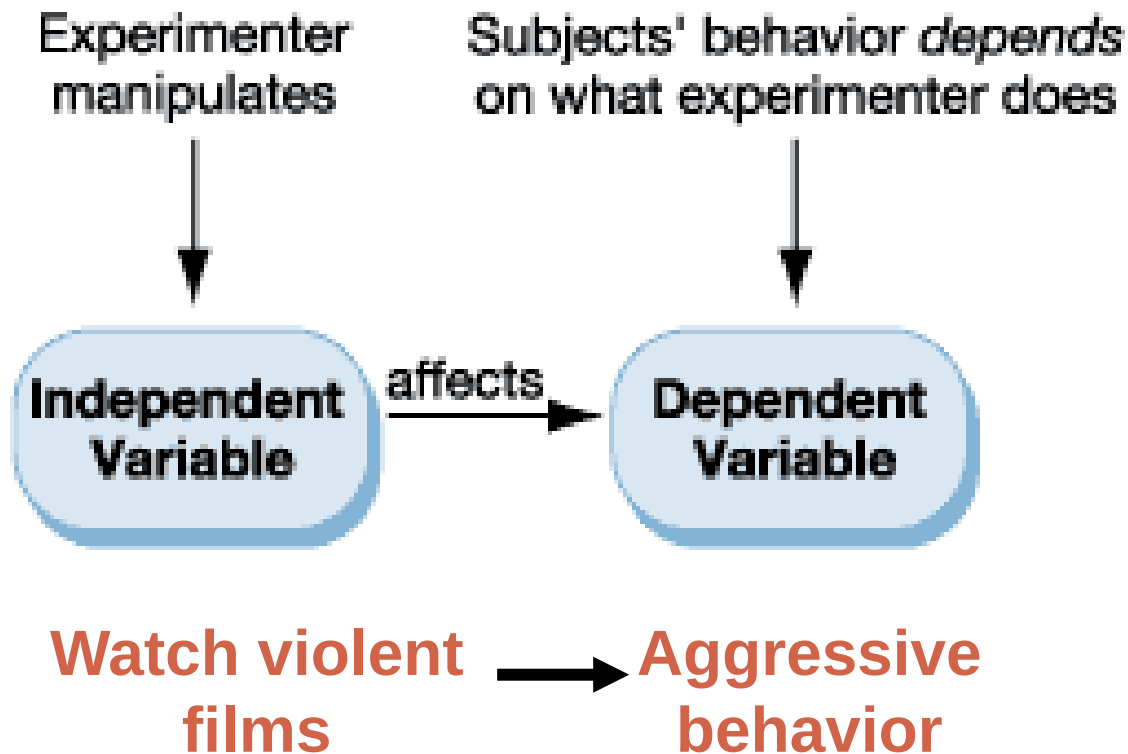
Goals of Psychology

To **explain** and **understand** behavior and mental processes

- **Experimentation**

- Like other sciences, experimentation is the backbone of psychological research.
- Experiments isolate causes and their effects.

Experimentation



Basic knowledge

- How do we know whether these claims are good?
 - **Validity**
 - the appropriateness of a claim
 - a valid claim is
 - Reasonable
 - Accurate
 - Justifiable

Construct validity

- An indication of how well a conceptual variable is measured or manipulated in a study
- How well is a construct operationalized?



Which Mona Lisa do you find more attractive?



0

1

0

2

Construct validity

- *For example:*
 - *what is the influence of familiarity on attractiveness?*



Construct validity

- *For example: what is the influence of familiarity on attractiveness?*

constructs: familiarity - attractiveness



operational variables: orientation - score

Construct validity

- Threats to construct validity
 - Inadequate operational definition
 - *e.g., does a test measure anxiety or absence of self-confidence?*
 - Mono-operation bias
 - *e.g., mental workload*

Construct validity

A researcher is interested in the influence of culture on the ability to solve complex problems.

He performs a quasi-experiment in which he measures how long people from different countries need to find their way to the baggage belts in Schiphol airport.

The study shows that Dutch travelers are generally faster to find their way in Schiphol airport than travelers from other countries.

The conclusion of his research is that the Dutch culture contributes to the ability to solve complex problems.

What can you tell about the construct validity in this study?

External validity

- An indication of how well the results of a study generalize to, or represent, individuals, settings, places, and times (contexts) besides those in the study itself
- Is it possible to generalize?

External validity



External validity

- Threats to external validity
 - Selection biases
 - *e.g., study on the effect of marijuana on the ability to solve math problems*

Sex and Gender: Critical Variables in Pre-Clinical and Clinical Medical Research

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In this Essay, we discuss the critical need to incorporate sex and gender in pre-clinical and clinical research to enhance our understanding of the mechanisms by which metabolic processes differ by sex and gender. This knowledge will allow for development of personalized medicine which will optimize therapies specific for individuals.

Most pre-clinical and clinical medical research, both animal and human, has been biased with respect to sex. There has been a tendency to treat the sexes as equivalent and not consider how fluctuations of sex hormones in experimental settings impact outcomes, thereby limiting our understanding of the molecular mechanisms that drive sexual dimorphisms. The National Institutes of Health (NIH) has recently recognized this gap in scientific knowledge and now mandates that studies be conducted in both sexes.

External validity

- **Random Assignment**

- Assigning participants to experimental and control conditions by chance
- Minimizes pre-existing differences between those assigned to the different groups

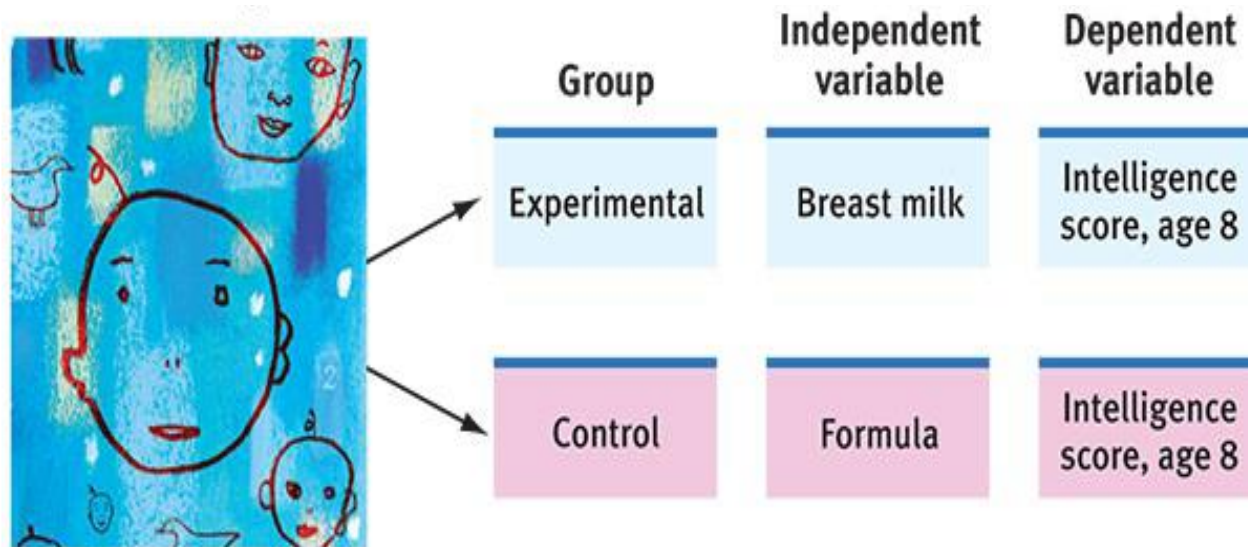
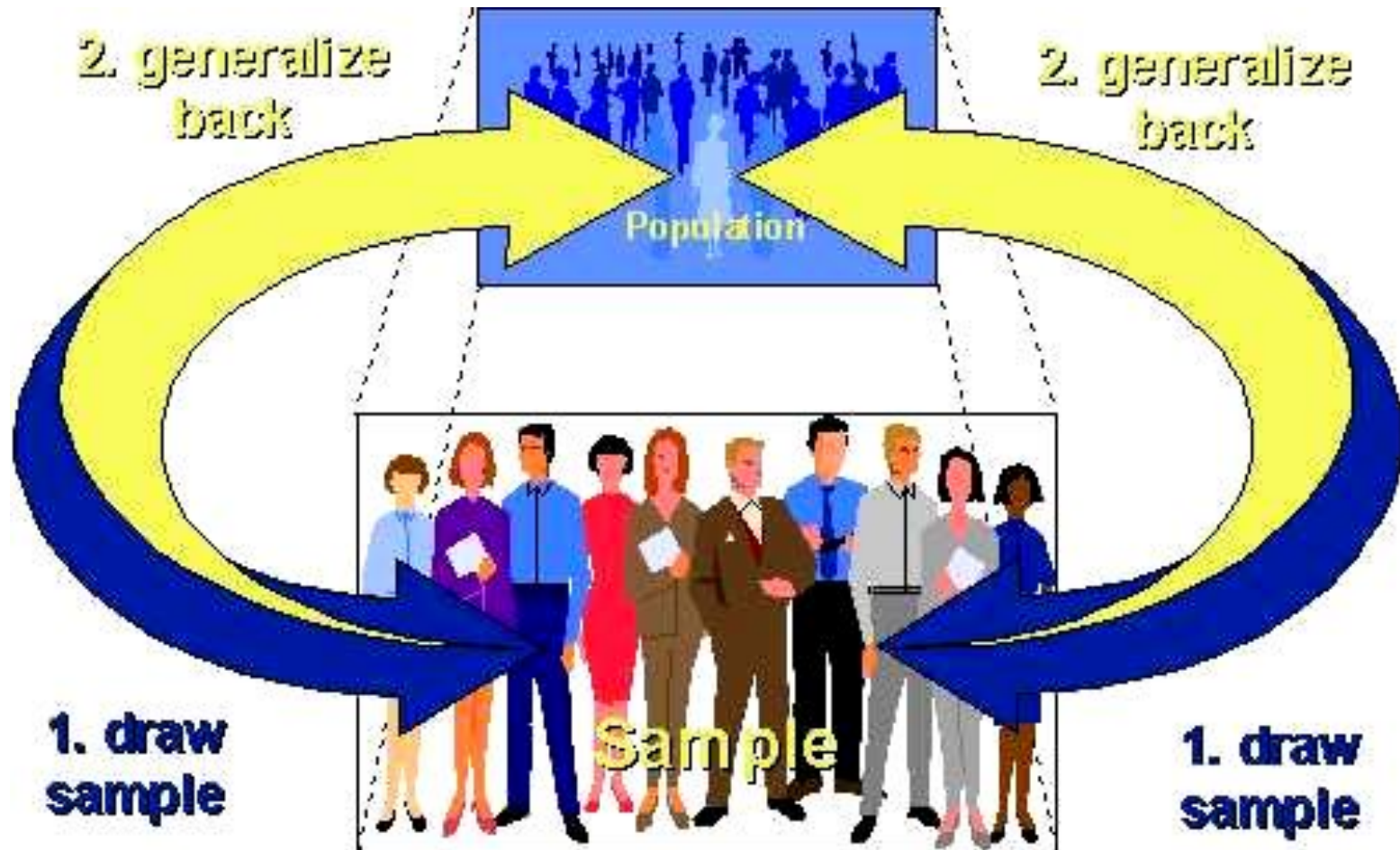


FIGURE 1.5 Myers/DeWall, *Psychology*, 12e, © 2018 Worth Publishers
R. Michael Wertz

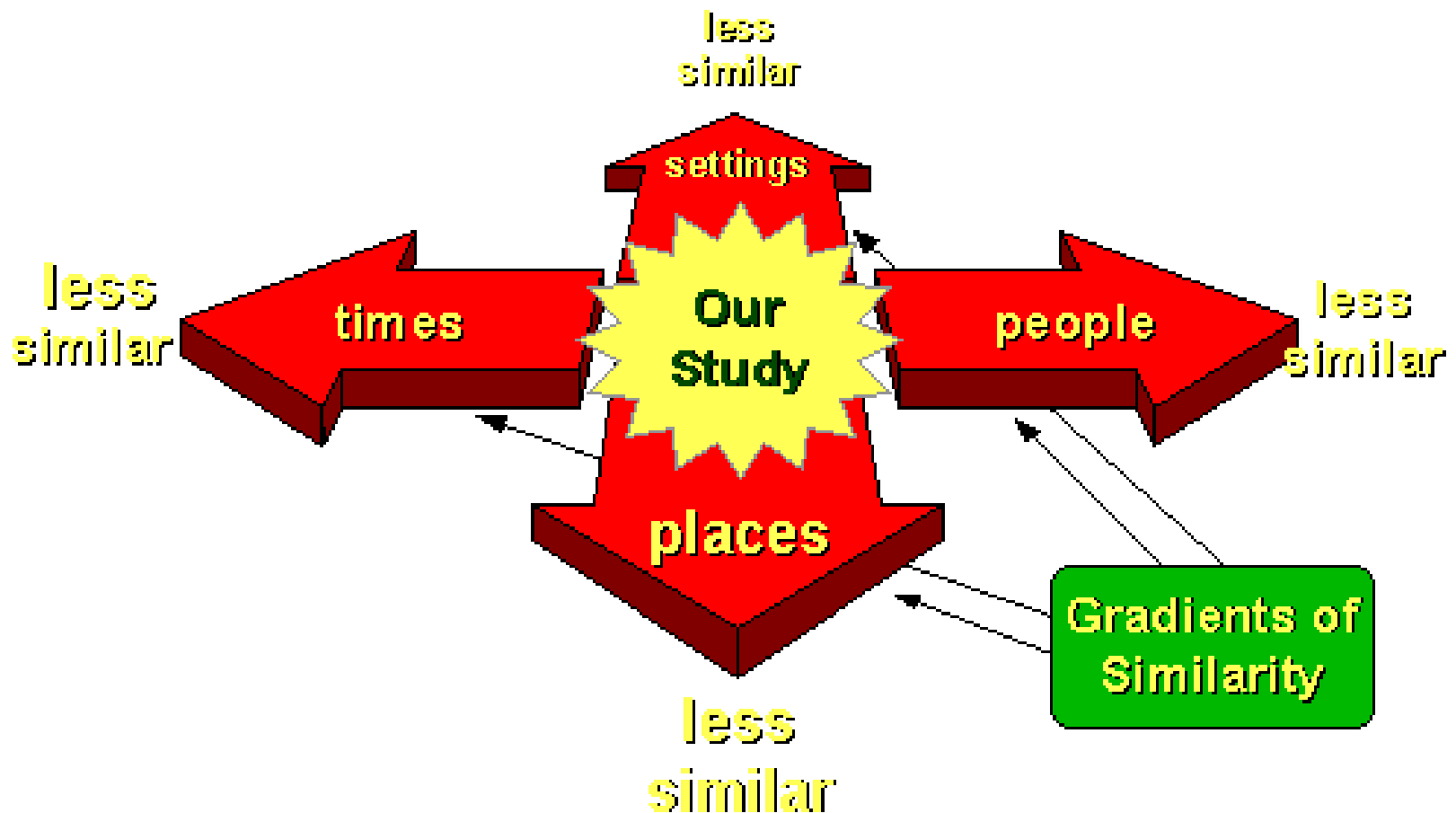
External validity



External validity

- Threats to external validity
 - Selection biases
 - *e.g., study on the effect of marijuana on the ability to solve math problems*
 - Study setting is different from other settings
 - *e.g., survey study on attitudes towards drug users*
 - *e.g., study on memory in a stuffy classroom*
 - *e.g., covid spread in urban vs. rural areas*
 - Specific time at which the study is performed is different from other times
 - *e.g., study on attitude concerning eating meat after BSE scandal*

External validity



External validity

A traffic psychologist is asked to test which of several alternative signs for "carpool lane" is best understood.

He performs an experiment in which he presents the various alternative signs to a group of students.

The students have the task to indicate which sign is the best sign for carpool lane.

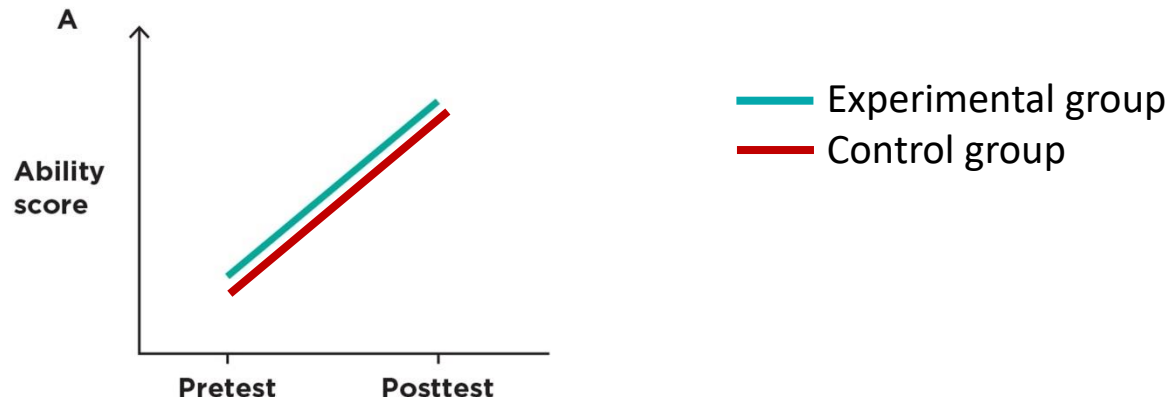
What can you tell about the external validity of this study?

Internal validity

- The extent to which it is possible to rule out alternative explanations for a causal relationship between two variables.
- Can we rule out alternative explanations?

Internal validity

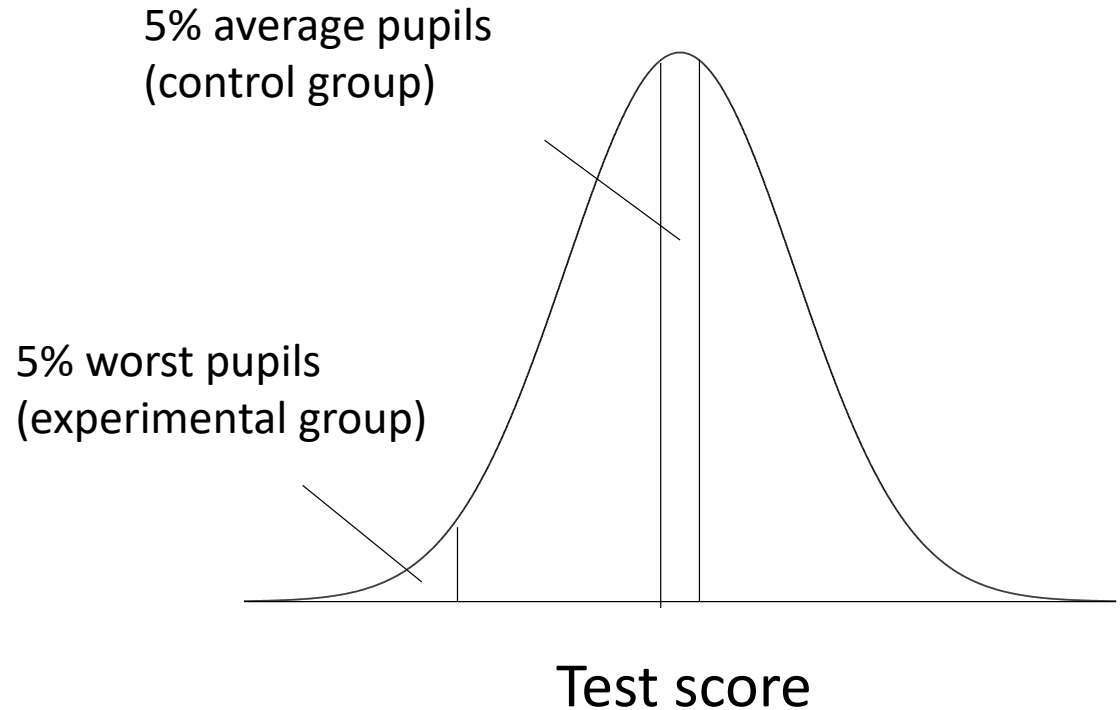
- Threats to internal validity
 - Maturation
 - History
 - Testing
 - Instrumentation
 - *e.g., study on the effect of a new educational math program on math ability*



Internal validity

- Threats to internal validity
 - Regression to the mean
 - Expected return to normal
 - *e.g., study of a new learning method in low- and medium-ranking pupils*

Internal validity



Score on test before the introduction of the new learning method

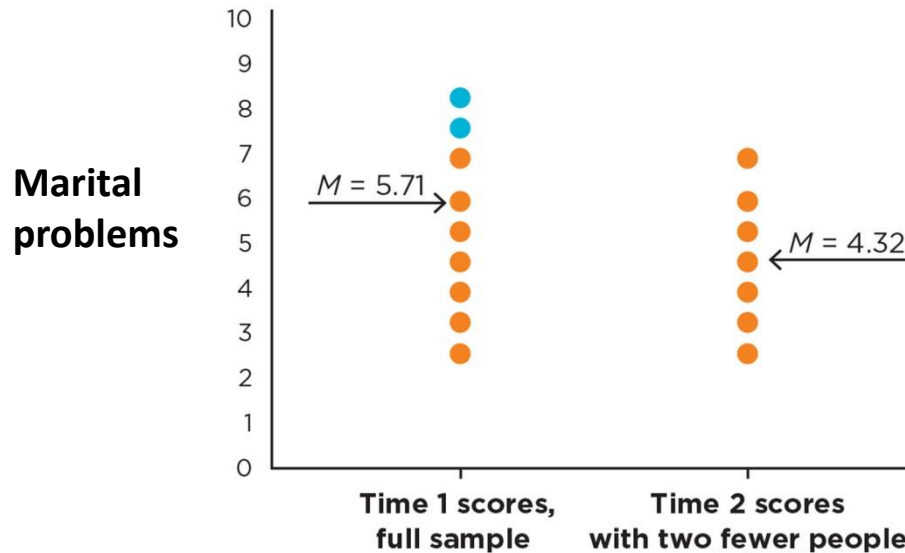
experimental group	40
control group	80

Score on test after the introduction of the new learning method

experimental group	50
control group	80

Internal validity

- Threats to internal validity
 - Attrition / Experimental Mortality
 - *e.g., study on the effect of couple counseling on the divorce rate*



Internal validity

- Threats to internal validity
 - Selection effects
 - *e.g., Simpson paradox*

Internal validity

Simpson paradox

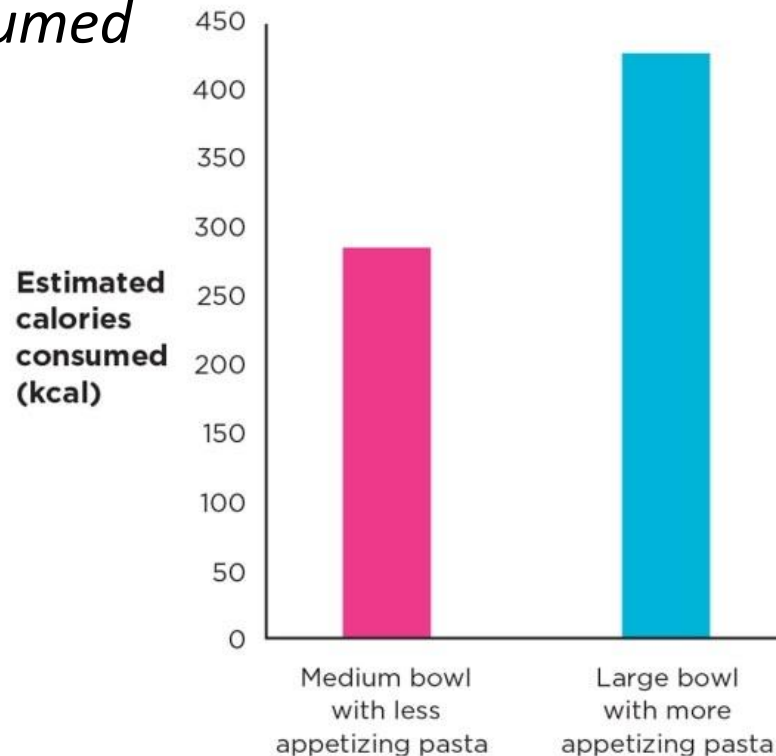
Internal validity

Simpson paradox

	no headache (recovered)		headache (not recovered)	% cured
N=80				
pill (N=40)	20		20	50%
no pill (N=40)	16		24	40%
men				
pill (N=30)	18	➤	12	60%
no pill (N=10)	7	➤	3	70%
women				
pill (N=10)	2	➤	8	20%
no pill (N=30)	9	➤	21	30%

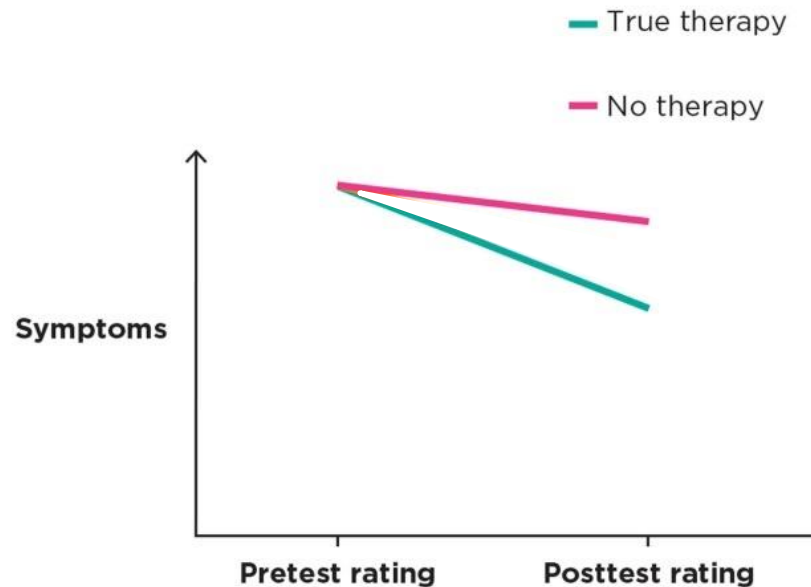
Internal validity

- Threats to internal validity
 - Design confounds
 - the presence of another variable that unintentionally varies systematically with the independent variable
 - *e.g., order effect like in the Stroop demonstration*
 - *e.g., study on the effect of bowl-size on estimates calories consumed*



Internal validity

- Threats to internal validity
 - Observer bias
 - *e.g., study on a new antidepressant*
 - Single- and double-blind studies
 - Demand characteristics
 - Placebo effects



Internal validity

A manufacturer of skin care products brings a new day cream on the market.

He claims that this day cream leads to significant skin improvement in women with skin problems.

He justifies his claims based on a study using a sample of 50 women with a problematic skin and 50 women with a normal skin.

The purity of the skin was precisely measured in both groups at the beginning of the study (using a purity index).

Internal validity

Following both groups daily applied the new day cream for two weeks. The results :

	Average purity index	
	Before treatment	After treatment
Group with problem skin	40	60
Group with normal skin	80	80

The results show a clear improvement in the condition of the skin after treatment compared to before in the group with skin problems. There were no effects in the group with normal skin.

The conclusion of the study was that daily use of the day cream has a positive impact on the quality of a problematic skin.

What can you say about the internal validity of this study?

Statistical validity

- The extent to which a study's statistical conclusions are accurate and reasonable
- How well do the numbers support the claims?

Statistical validity

- Threats to statistical validity
 - Violated assumptions of the test statistics
 - Fishing and the error rate problem
 - *Seeing things that aren't there*
 - Low statistical power
 - *Missing the needle in the haystack*
 - Unreliability of measures
- for more information:
see your statistics course

